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(54) **LIGHT EMITTING DIODE PACKAGE,
DIVISIONAL DISPLAY MODULE, AND
DISPLAY PANEL**

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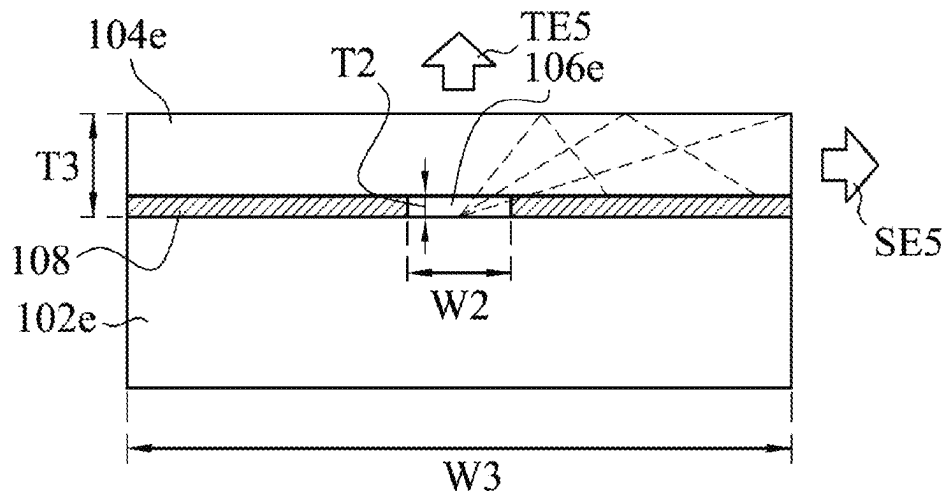
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ABSTRACT

A light emitting diode (LED) package includes a substrate,
at least one micro LED chip, a black material layer, and a
transparent material layer. The substrate has a width ranging
from 100 micrometers to 1000 micrometers. The at least one
micro LED chip is electrically mounted on a top surface of
the substrate and has a width ranging from 1 micrometer to
100 micrometers. The black material layer covers the top
surface of the substrate to expose the at least one micro LED
chip. The transparent material layer covers the at least one
micro LED chip and the black material layer.

20 Claims, 12 Drawing Sheets

100e



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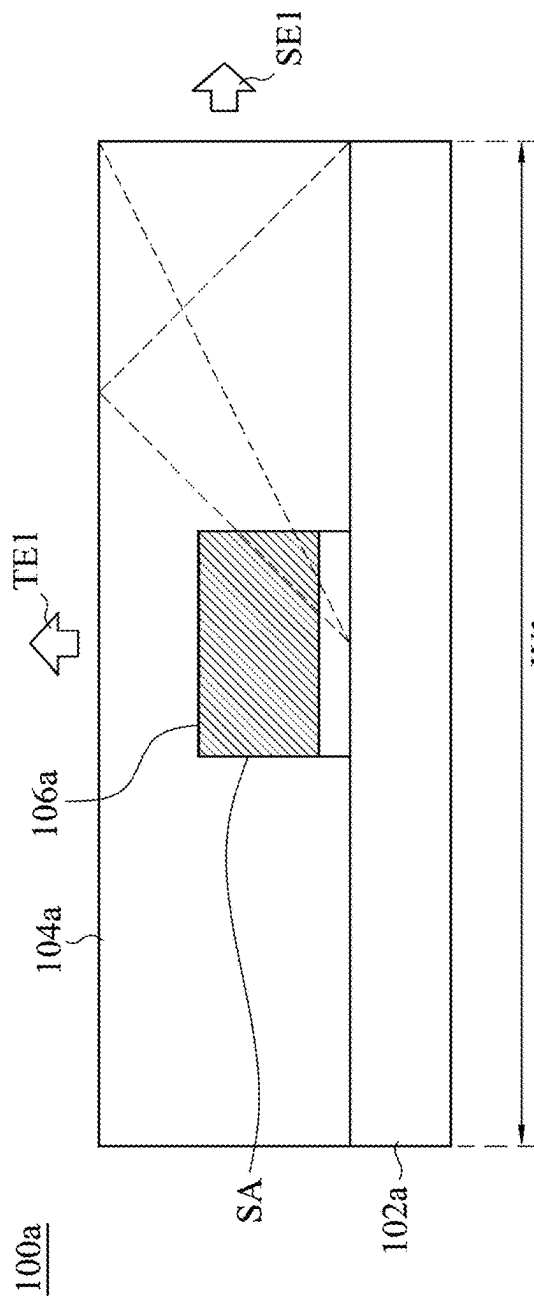


Fig. 1
(PRIOR ART)

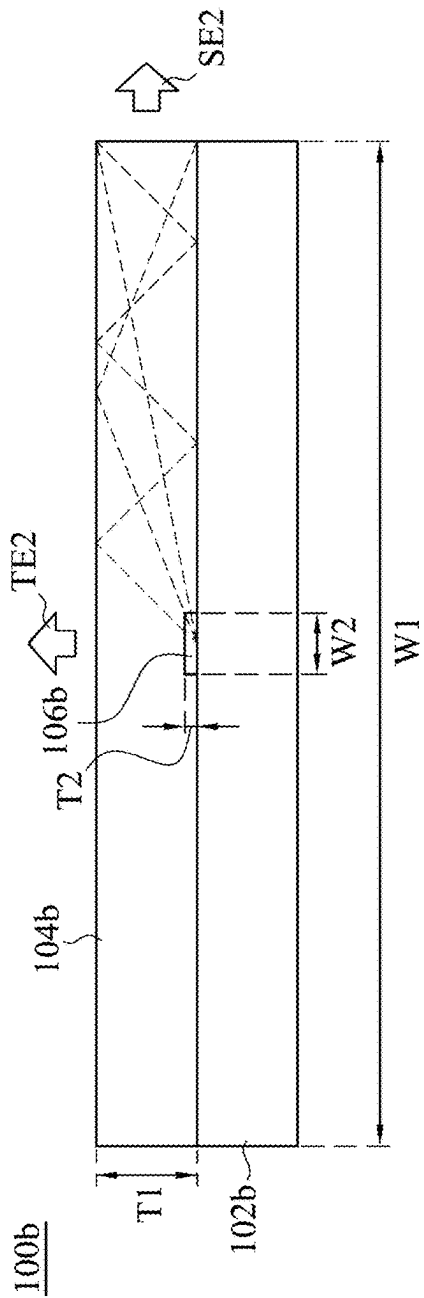


Fig. 2

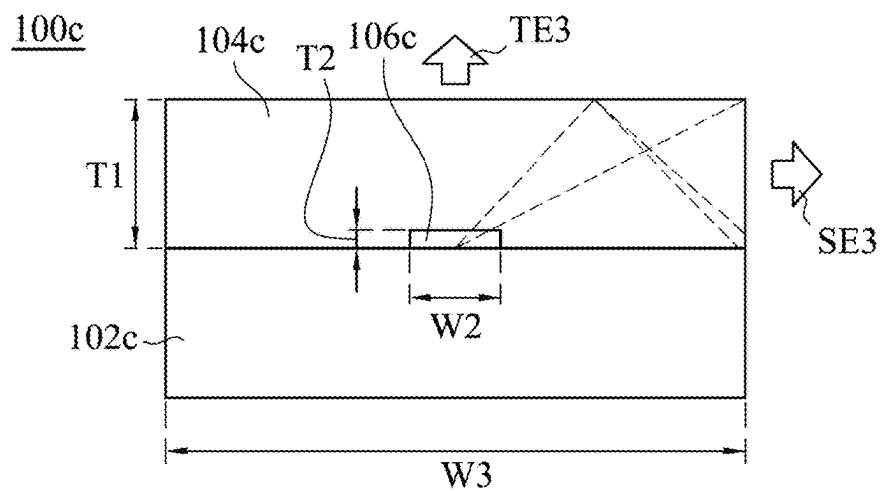


Fig. 3

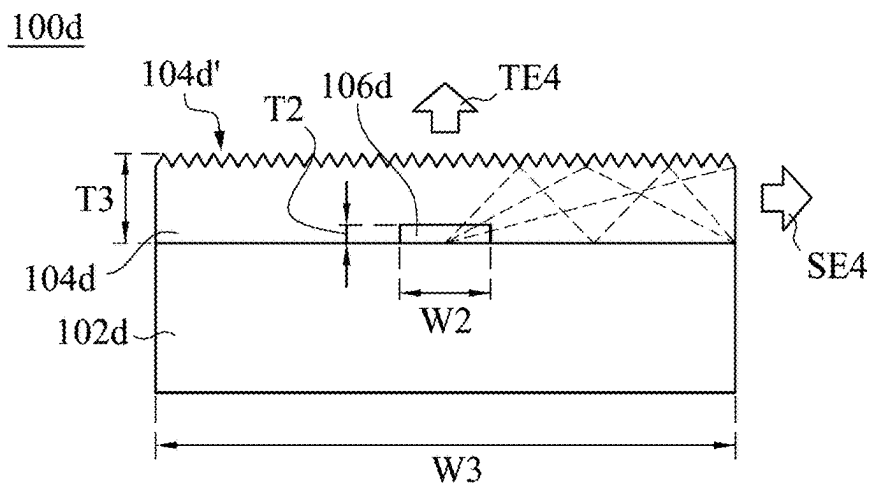


Fig. 4

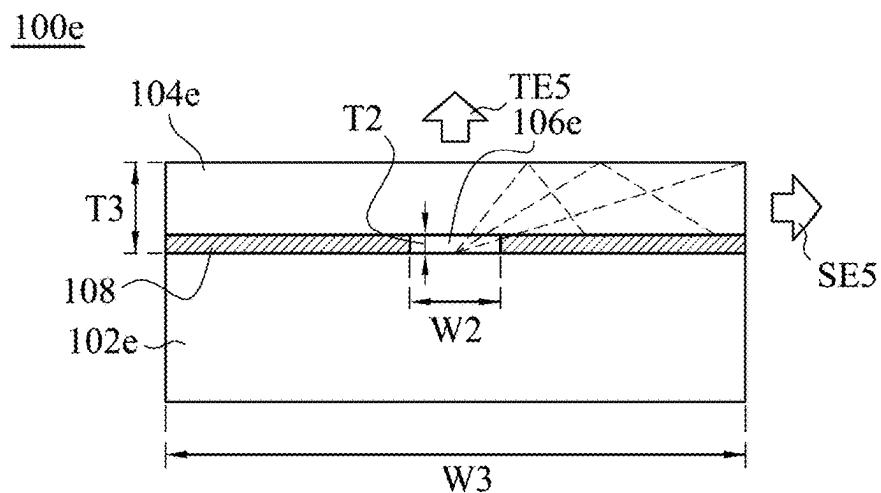


Fig. 5

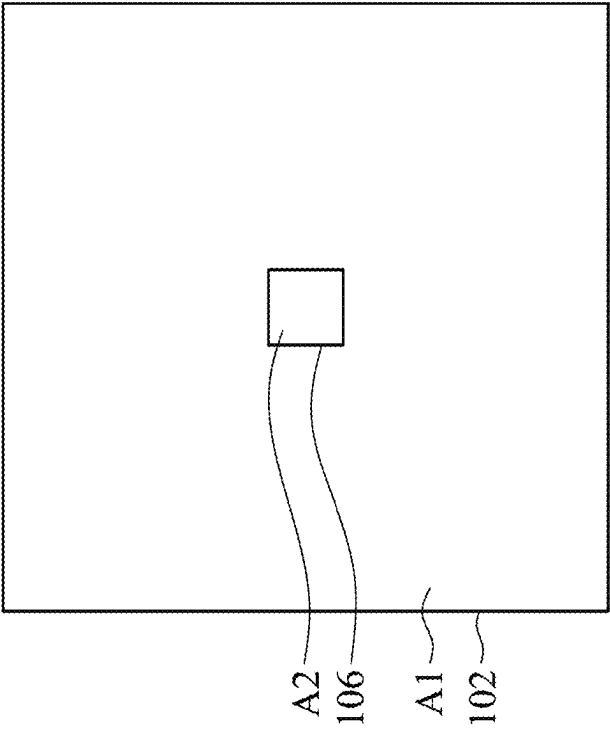


Fig. 6

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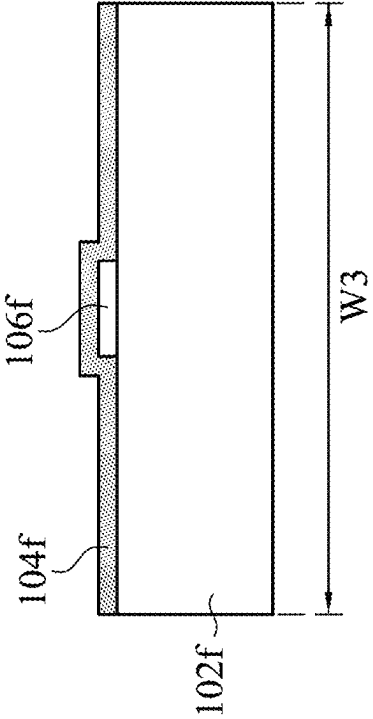


Fig. 7

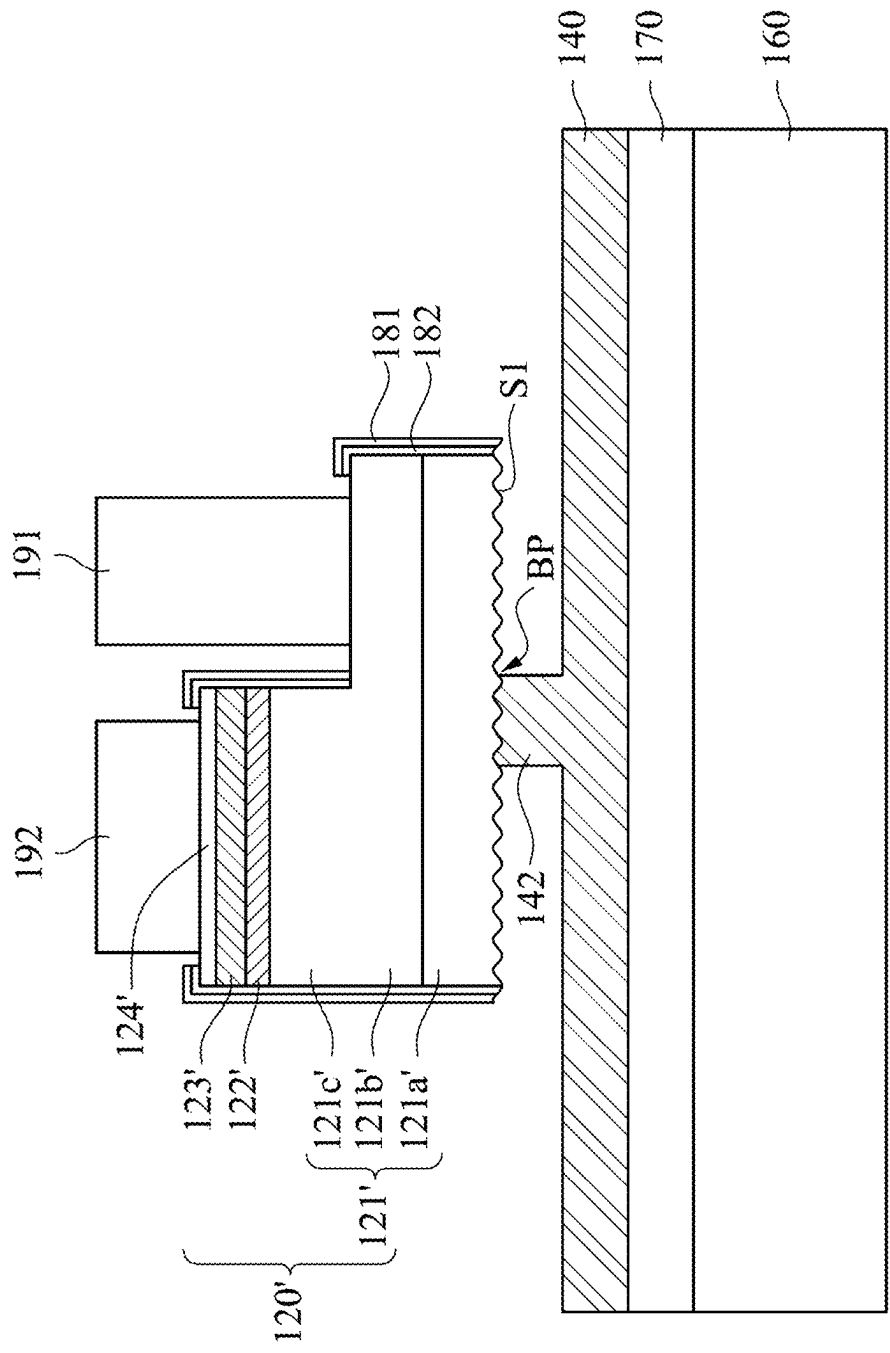
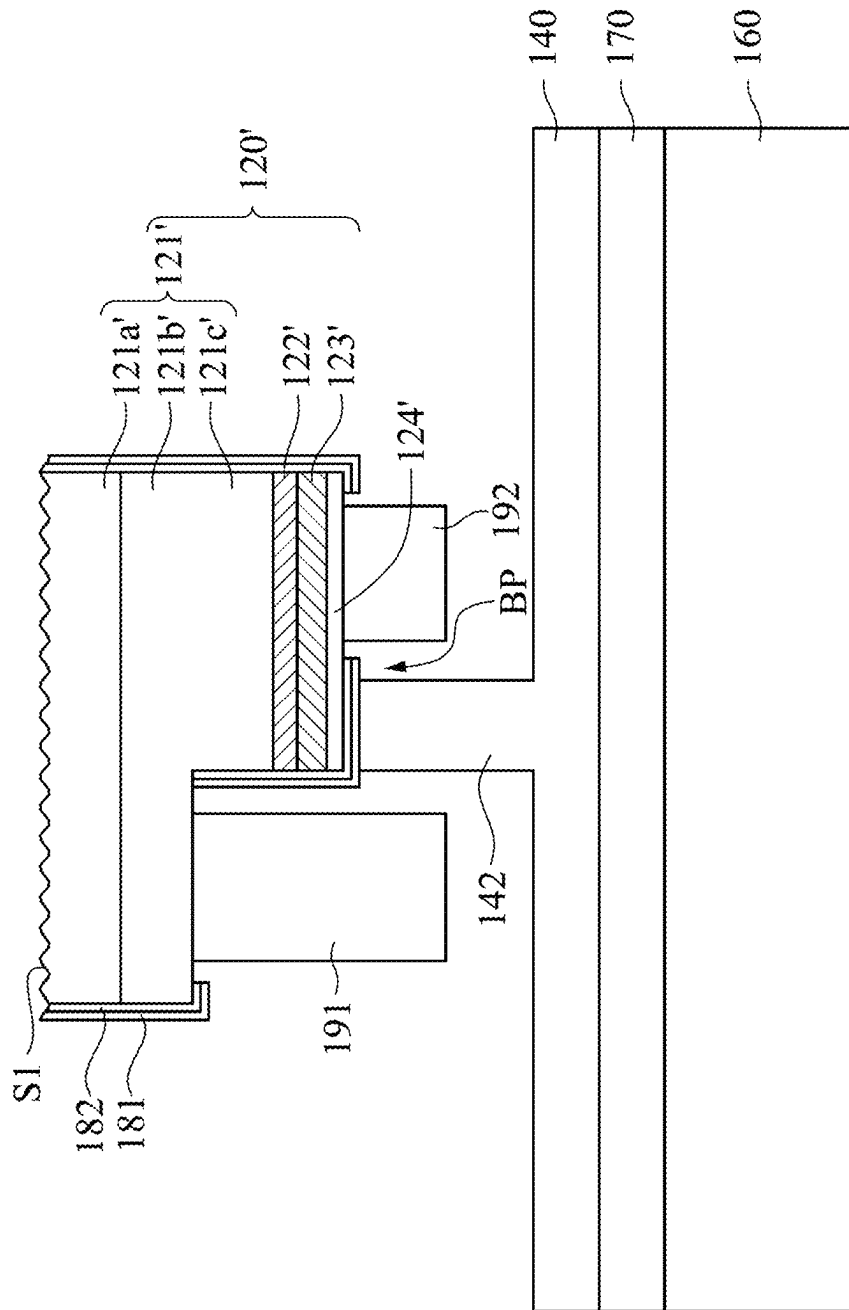
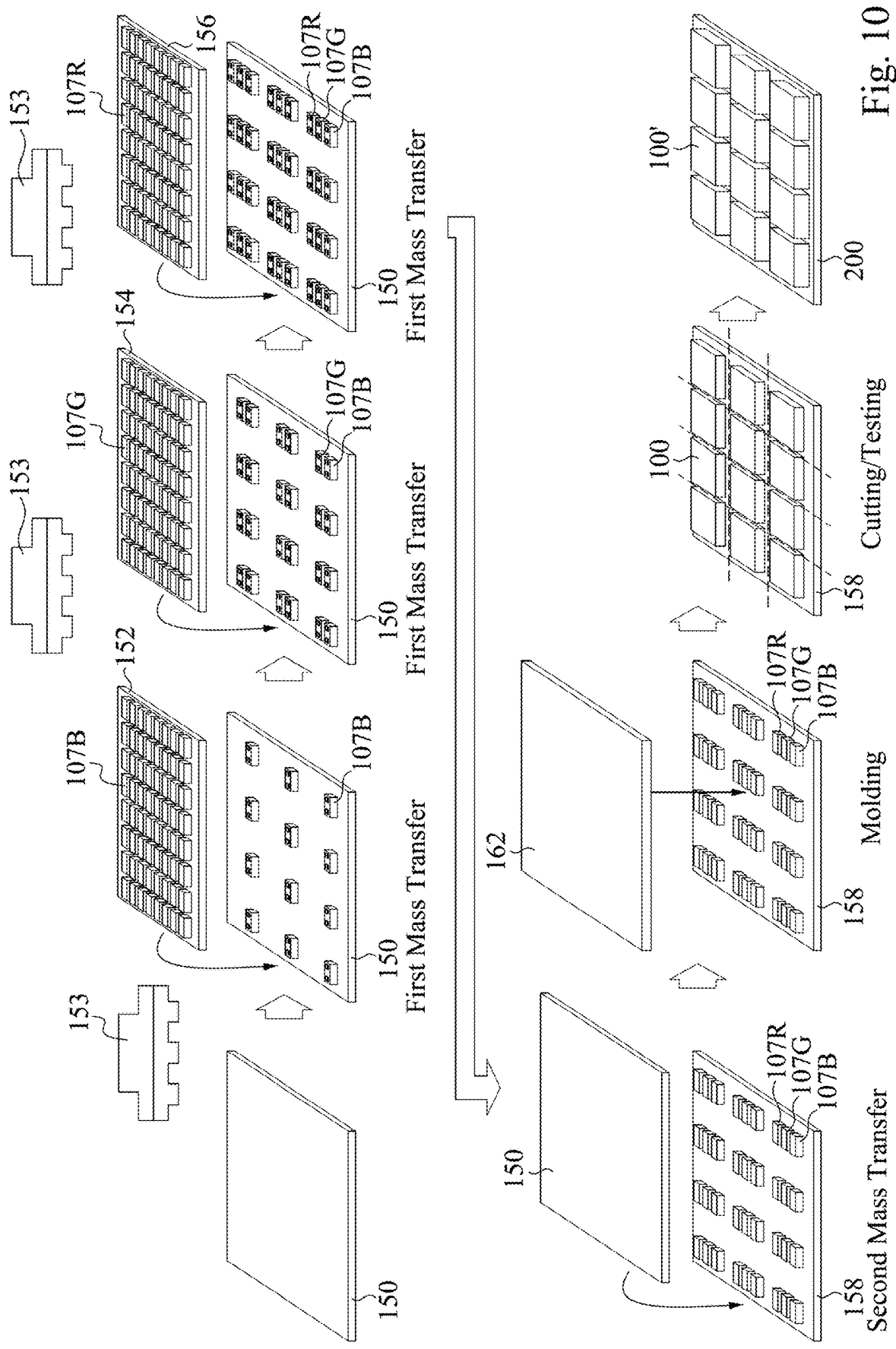
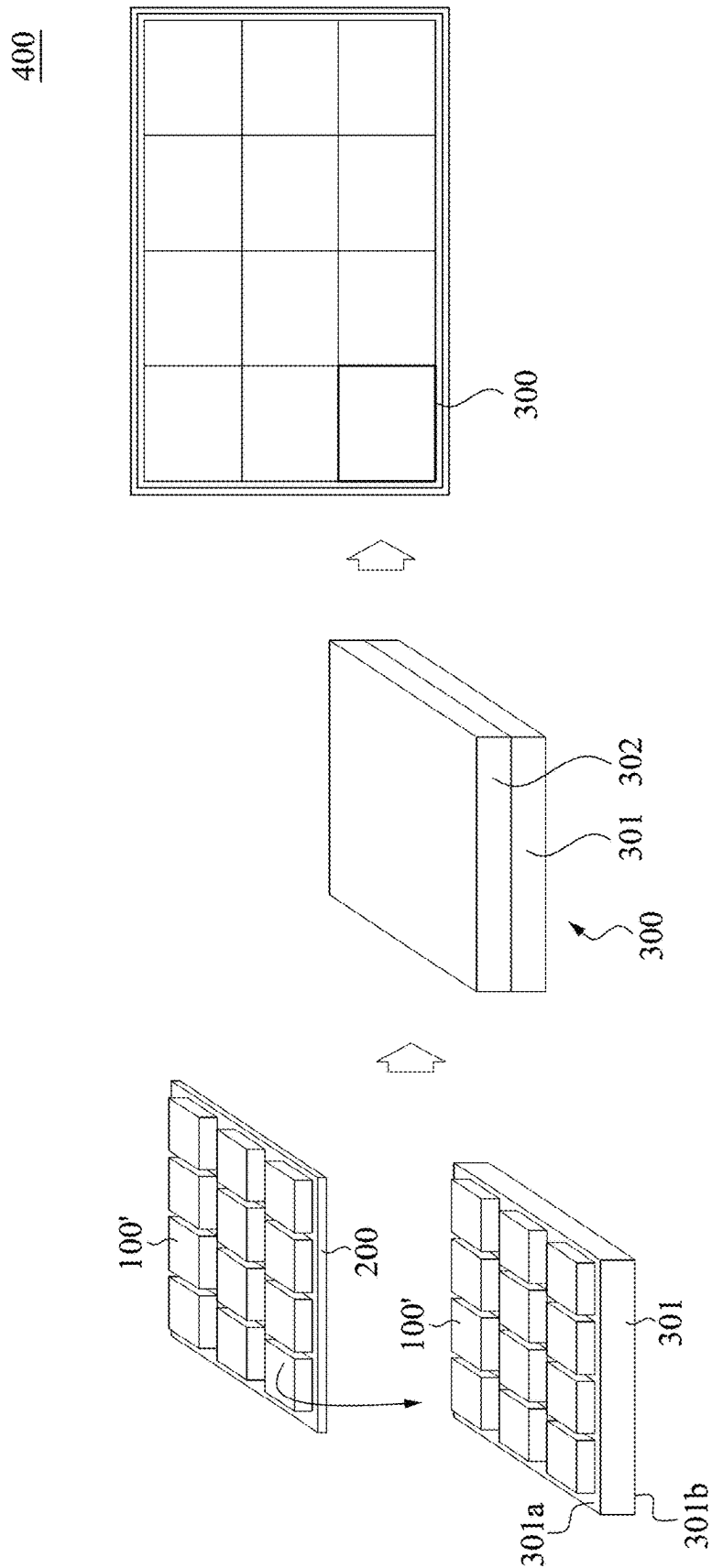


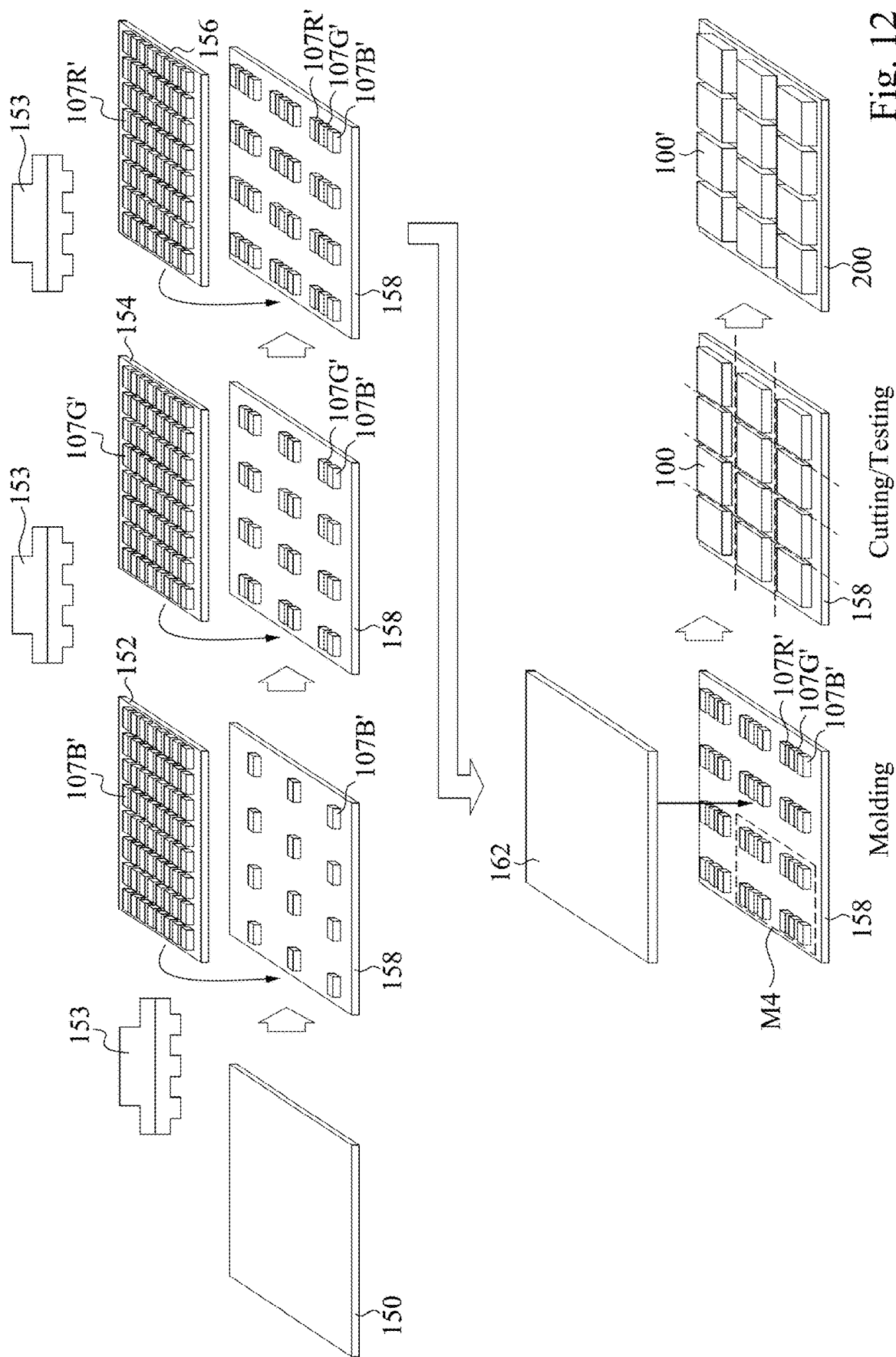
Fig. 8

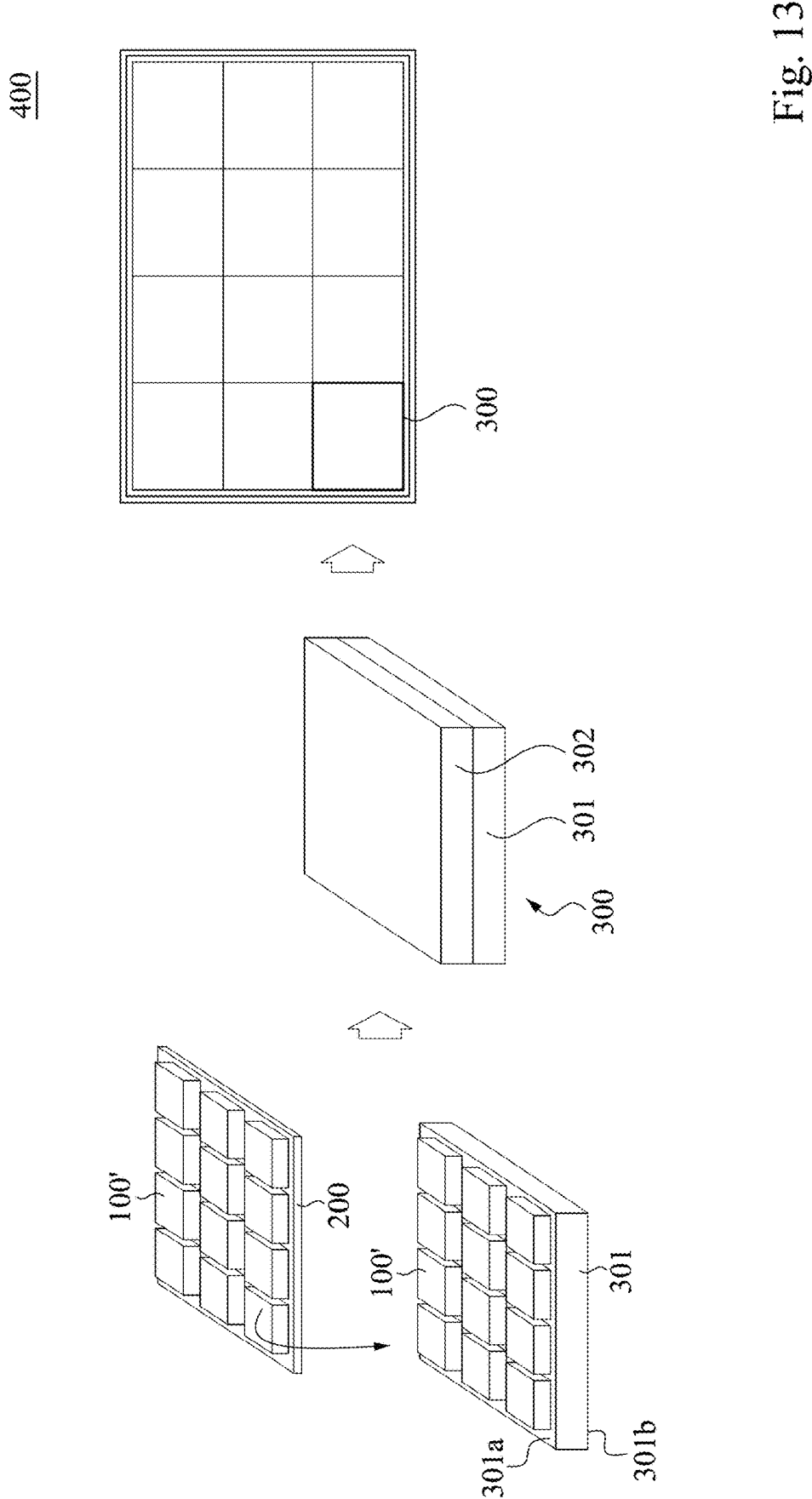
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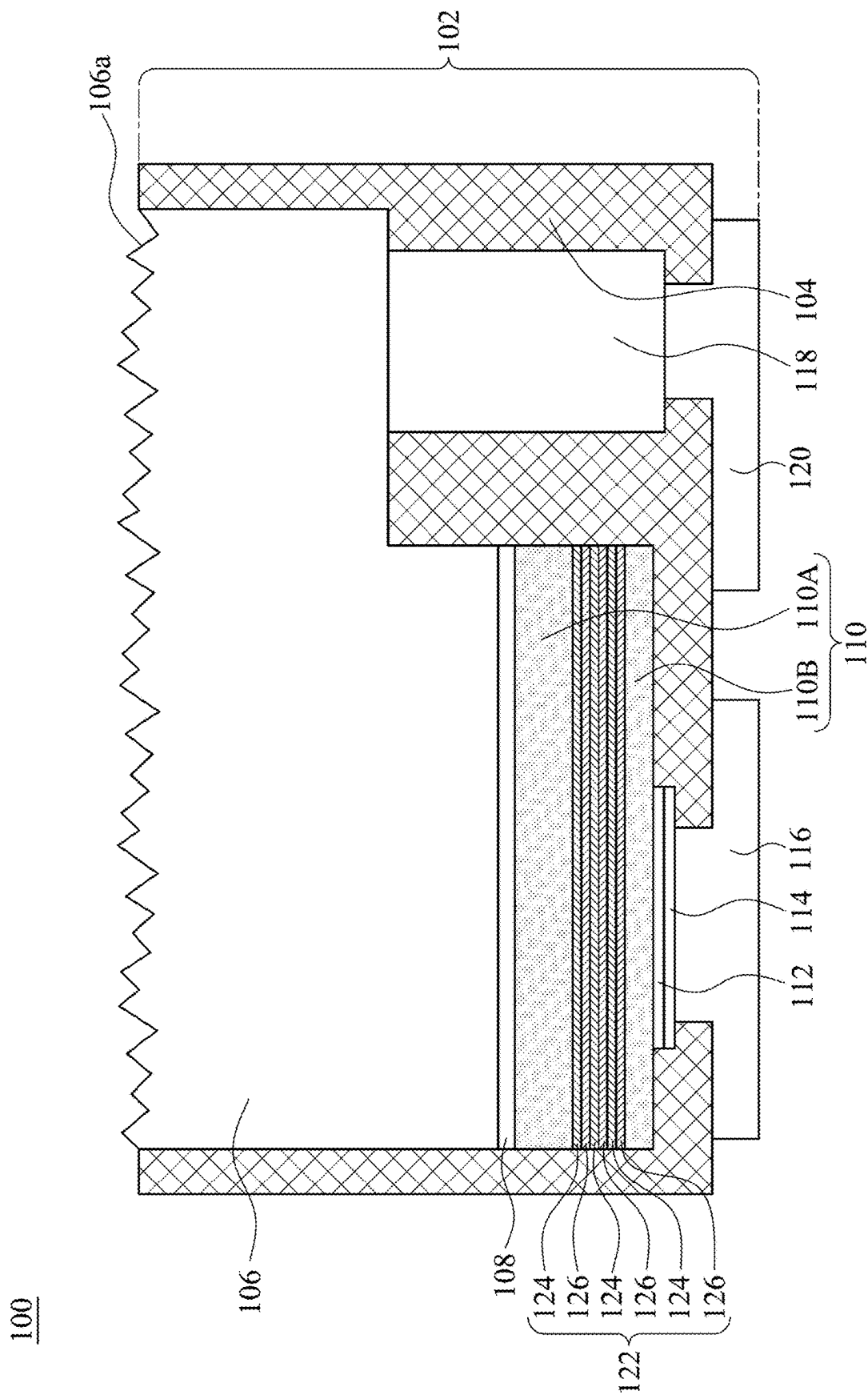


Fig. 14

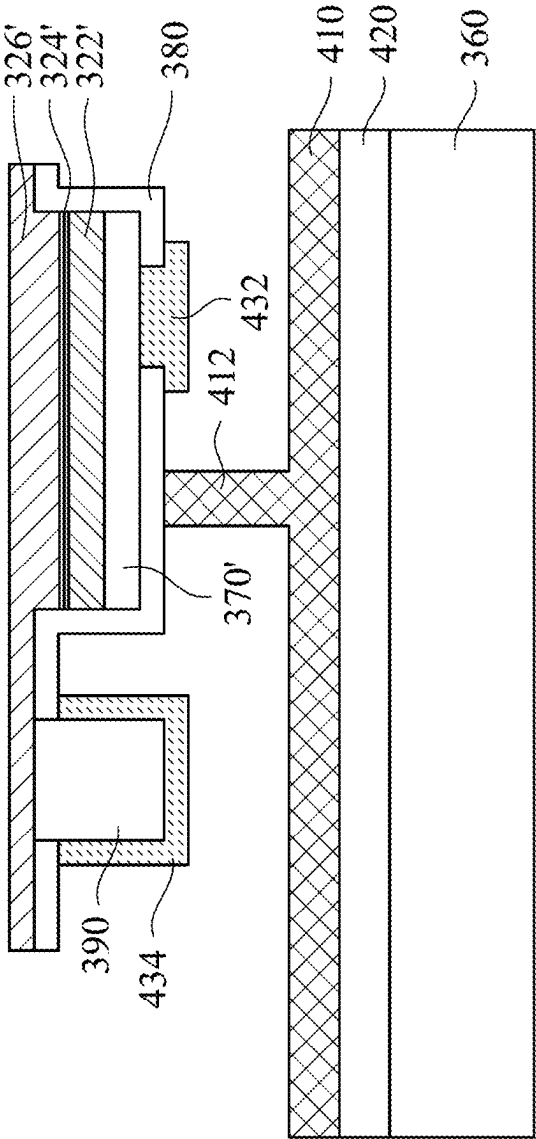


Fig. 15

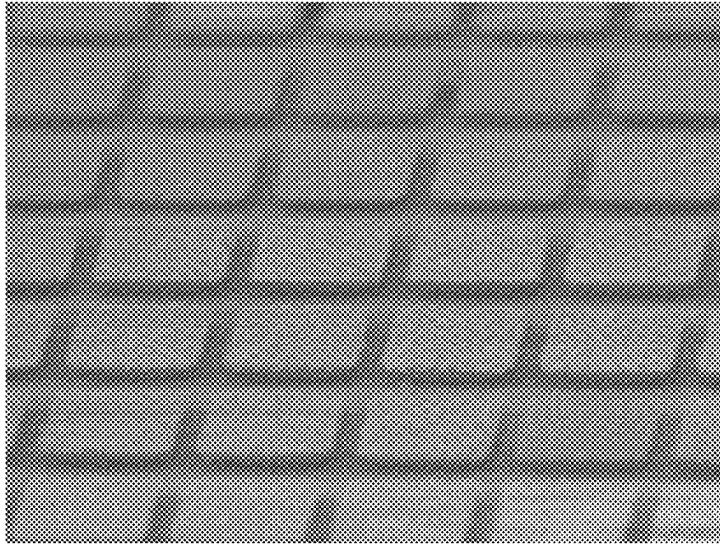


Fig. 16

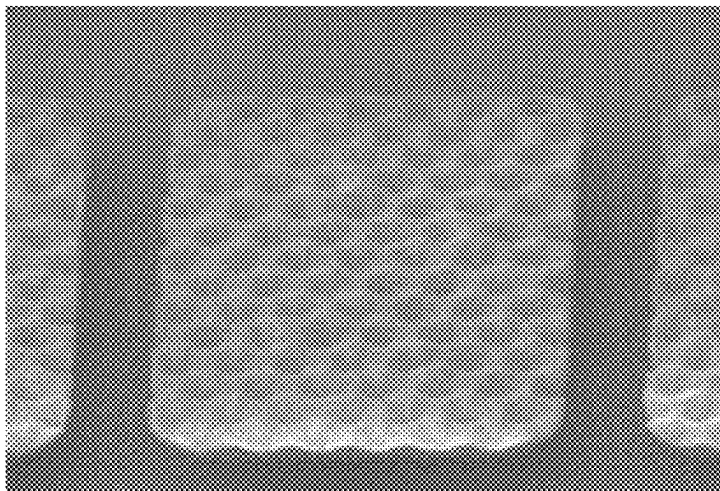


Fig. 17

1

LIGHT EMITTING DIODE PACKAGE, DIVISIONAL DISPLAY MODULE, AND DISPLAY PANEL

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 18/172,283, filed on Feb. 21, 2023, which is a continuation of U.S. application Ser. No. 17/308,070, filed on May 5, 2021, now U.S. Pat. No. 11,616,173, issued Mar. 28, 2023, which is a continuation of U.S. application Ser. No. 16/600,577 filed on Oct. 14, 2019, now U.S. Pat. No. 11,038,088, issued Jun. 15, 2021, which is related to copending U.S. application Ser. No. 16/541,132 filed on Aug. 14, 2019, copending U.S. application Ser. No. 16/524,165 filed on Jul. 29, 2019 and copending U.S. application Ser. No. 16/524,202 filed on Jul. 29, 2019, all of which are incorporated by reference herein in their entireties and not admitted to be prior art with respect to the present disclosure by their mention in this cross-reference section.

BACKGROUND

Field of Invention

The present disclosure relates to a light emitting diode package.

Description of Related Art

It is known to use LED displays each including an array of red, green, and blue LED elements as pixels. Such LED displays offer light with higher luminance than the luminance of backlight-type liquid-crystal displays and are used for large-scale digital signage and other applications.

There is a demand for such LED displays to have high contrast ratios, but reflection of extraneous light may cause false lighting and reduce the contrast ratios in environments.

Therefore, there is a need to provide a light-emitting device that offers higher luminance and a higher contrast ratio.

SUMMARY

One aspect of the present disclosure provides a light emitting diode package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate has a width ranging from 100 micrometers to 1000 micrometers. The plurality of micro LED chips are electrically mounted on a top surface of the substrate and has a width ranging from 1 micrometer to 100 micrometers. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. Each micro LED chip has a thickness substantially equal to that of the black material layer. The transparent material layer covers the plurality of micro LED chips and the black material layer. The micro LED chip includes a first semiconductor layer with a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer;

2

and a supporting breakpoint on the light emitting surface. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

Another aspect of the present disclosure provides a light emitting diode (LED) package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate having a width ranging from 100 micrometers to 1000 micrometers. The plurality of micro LED chips are electrically mounted on a top surface of the substrate and has a width ranging from 1 micrometer to 100 micrometers. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. Each micro LED chip has a thickness substantially equal to that of the black material layer. The transparent material layer covers the micro LED chips and the black material layer. The transparent material layer has a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

Another aspect of the present disclosure provides a light emitting diode (LED) package. The light emitting diode package includes a substrate, a plurality of micro LED chips, a black material layer, a transparent material layer, and a pixel controller. The substrate having a width. The plurality of micro LED chips is electrically mounted on a top surface of the substrate. The plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip. The black material layer covers the top surface of the substrate to expose the plurality of micro LED chips. The plurality of micro LED chips has a thickness substantially equal to that of the black material layer. The transparent material layer covers the micro LED chips. The transparent material layer has a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4. The micro LED chip includes a first semiconductor layer with a light emitting surface exposed outside and the light emitting surface has a rough texture; a light emitting layer disposed on the first semiconductor layer; a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and a supporting breakpoint on the light emitting surface. The pixel controller controls the blue micro LED chip, the green micro LED chip, and the red micro LED chip.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1 illustrates a cross-sectional view of a typical light emitting diode package.

FIGS. 2-5 illustrate cross-sectional views of light emitting diode packages in accordance with embodiments of the present invention.

FIG. 6 illustrates a top view of the light emitting diode package in accordance with another embodiment of the present invention.

FIG. 7 illustrates a cross-sectional view of a light emitting diode package in accordance with another embodiment of the present invention.

3

FIG. 8 illustrates a cross-sectional view of a micro LED chip in accordance with another embodiment of the present invention.

FIG. 9 illustrates a cross-sectional view of a micro LED chip in accordance with still another embodiment of the present invention.

FIGS. 10-11 illustrate steps of manufacturing a light emitting diode display panel in accordance with one embodiment of the present invention.

FIGS. 12-13 illustrate steps of manufacturing a light emitting diode display panel in accordance with another embodiment of the present invention.

FIG. 14 illustrates a cross-sectional view of a light emitting device according in accordance with another embodiment of the present invention.

FIG. 15 illustrates a cross-sectional view of a light emitting diode structure according in accordance with another embodiment of the present invention.

FIG. 16 is a photo of micro LED chip arrays of which each chip is equipped with a laser-lift-off rough pattern.

FIG. 17 is a photo of a top surface of a micro LED chip equipped with a laser-lift-off rough pattern.

DETAILED DESCRIPTION

The present disclosure is described by the following specific embodiments. Those with ordinary skill in the arts can readily understand the other advantages and functions of the present invention after reading the disclosure of this specification. The present disclosure can also be implemented with different embodiments. Various details described in this specification can be modified based on different viewpoints and applications without departing from the scope of the present disclosure.

The following embodiments are disclosed with accompanying diagrams for detailed description. For illustration clarity, many details of practice are explained in the following descriptions. However, it should be understood that these details of practice do not intend to limit the present invention. That is, these details of practice are not necessary in parts of embodiments of the present invention. Furthermore, for simplifying the drawings, some of the conventional structures and elements are shown with schematic illustrations.

FIG. 1 illustrates a cross-sectional view of a typical light emitting diode package. A LED package 100a includes a substrate 102a, at least one mini LED chip 106a and a transparent material layer 104a. The mini LED chip 106a conventionally includes a sapphire substrate SA. The transparent material layer 104a covers the at least one mini LED chip 106a and the substrate 102a. In this embodiment, the substrate 102a has a width W1 ranging from 100 micrometers to 1000 micrometers. The mini LED chip 106a has a width ranging from 100 micrometers to 250 micrometers.

In an embodiment A of a package width of 1000 micrometers (i.e., the substrate 102a has a width of 1000 micrometers), a LED chip 106a with a thickness of 150 micrometers and a width of 225 micrometers, and a transparent glue layer (e.g., 104a) with a thickness equal to or greater than 250 micrometers, but no black material is covered over the substrate 102a, the LED chip 106a may emit to enable the package 100a to achieve a 18% ratio of a side emission SE1 to a top emission TE1. In this embodiment, a ratio of the width (W1) of the substrate 102a to a thickness of the transparent material layer 104a is smaller than 4.

As shown in FIG. 2, a LED package 100b includes a substrate 102b, at least one micro LED chip 106b and a

4

transparent material layer 104b. In this embodiment, the substrate 102b has a width W1 ranging from 100 micrometers to 1000 micrometers. The substrate 102b is used to package a micro LED chip 106b having a width W2 ranging from 1 micrometers to 100 micrometers and a thickness T2 smaller than 10 micrometers, instead of a mini LED chip, e.g., 106a. The transparent material layer 104b covers the at least one micro LED chip 106b and the substrate 102b.

In certain embodiments, the substrate 102b has a width W1 ranging from 100 to 200 micrometers, from 200 to 500 micrometers, from 500 to 750 micrometers, or from 750 to 1000 micrometers.

In certain embodiments, the micro LED chip 106b has a width W2 ranging from 1 micrometer to 100 micrometers, e.g., from 1 to 5 micrometers, from 5 to 10 micrometers, from 10 to 25 micrometers, or from 25 to 50 micrometers.

In an embodiment B of a package width of 1000 micrometers (i.e., the substrate 102b has a width of 1000 micrometers), a LED chip 106b with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., 104b) with a thickness T1 of 100 micrometers, but no black material is covered over the substrate 102b, the LED chip 106b may emit to enable the package 100b to achieve a 5% ratio of a side emission SE2 to a top emission TE2.

Comparing the embodiments A and B, the side emission in embodiment B is reduced because the micro LED chip 106b is used to replace the mini LED chip 106a in the package and the transparent glue layer is downsized in its thickness, thereby increasing an internal reflection within the transparent glue layer.

As shown in FIG. 3, a LED package 100c includes a substrate 102c, at least one micro LED chip 106c and a transparent material layer 104c. In this embodiment, the substrate 102c has a width W3 ranging from 100 micrometers to 1000 micrometers. The substrate 102c is used to package a micro LED chip 106c having a width W2 ranging from 1 micrometers to 100 micrometers and a thickness T2 smaller than 10 micrometers. The transparent material layer 104c covers the at least one micro LED chip 106c and the substrate 102c.

In an embodiment C of a package width of 400 micrometers (i.e., the substrate 102c has a width of 400 micrometers), a LED chip 106c with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., 104c) with a thickness T1 of 100 micrometers, but no black material is covered over the substrate 102c, the LED chip 106c may emit to enable the package 100c to achieve a 10% ratio of a side emission SE3 to a top emission TE3.

Comparing the embodiments B and C, the side emission in embodiment C is increased because the package or the substrate 102c is downsized in its width to reduce an internal reflection within the transparent glue layer.

As shown in FIG. 4, a LED package 100d includes a substrate 102d, at least one micro LED chip 106d and a transparent material layer 104d. In this embodiment, the substrate 102d has a width W3 ranging from 100 micrometers to 1000 micrometers. The substrate 102d is used to package a micro LED chip 106d having a width ranging from 1 micrometer to 100 micrometers and a thickness T2 smaller than 10 micrometers. The transparent material layer 104d covers the at least one micro LED chip 106d and the substrate 102d.

In an embodiment D of a package width of 400 micrometers (i.e., the substrate 102d has a width of 400 micrometers), a LED chip 106d with a thickness of 10 micrometers and a width of 50 micrometers, and a transparent glue layer (e.g., 104d) with a thickness T3 of 50 micrometers, but no

black material is covered over the substrate **102d**, the LED chip **106d** may emit to enable the package **100d** to achieve a 4% ratio of a side emission **SE4** to a top emission **TE4**.

Comparing the embodiments C and D, the side emission in embodiment D is reduced because the transparent glue layer **104d** is downsized in its thickness to further increase an internal reflection within the transparent glue layer.

As shown in FIG. 4, the at least one micro LED chip **106e** has a thickness substantially equal to that of the black material layer **108**, but not being limited thereto.

In other embodiments, the transparent glue layer **104d** may have a top text surface **104d'** to further increase the top emission **TE4** for the LED package **100d**.

As shown in FIG. 5, a LED package **100e** includes a substrate **102e**, at least one micro LED chip **106e**, a black material layer **108** and a transparent material layer **104e**. In this embodiment, the substrate **102e** has a width **W3** ranging from 100 micrometers to 1000 micrometers. The substrate **102e** is used to package a micro LED chip **106e** having a width ranging from 1 micrometer to 100 micrometers and a thickness smaller than 10 micrometers. The black material layer **108** is configured to cover a top surface of the substrate **102e** and expose a light-emitting surface of the at least one micro LED chip **106a**. The black material layer **108** preferably has a thickness less than 10 micrometers.

In an embodiment E of a package width of 400 micrometers (i.e., the substrate **102e** has a width of 400 micrometers), a LED chip **106e** with a thickness of 10 micrometers and a width of 50 micrometers, a black material layer **108** with a thickness 3 micrometers and a reflectivity smaller than 10%, and a transparent glue layer (e.g., **104e**) with a thickness **T3** of 50 micrometers, the LED chip **106e** may emit to enable the package **100e** to achieve a 0.4% ratio of a side emission **SE5** to a top emission **TE5**. In this embodiment, the black material layer **108** may be a black photoresist having a reflectivity smaller than 10%, but not being limited thereto.

Comparing the embodiments D and E, the side emission in embodiment E is further reduced because the black material layer **108** is added to reduce internal reflection within the transparent material layer **104e**.

In embodiments B-E, the transparent material layer may have an optical transmittance greater than or equal to 90%, 92%, or 95%, and have a thickness smaller than 100 micrometers, but not being limited thereto.

In embodiments B-E, a ratio of the width (**W1**, **W3**) of the substrate (**102a-102e**) to the thickness (**T1**, **T3**) of the transparent material layer (**104a-104e**) is equal to or greater than 4 to suppress the side emission from the LED package such that the ratio of the side emission to the top emission can be reduced.

In embodiments B-E, a width of a LED chip may be referred as a longer edge of the LED chip or any edge of the LED chip in a square shape while a width of a substrate may be referred as a longer edge of the substrate or any edge of the substrate in a square shape, but not being limited thereto.

In embodiments B-E, the at least one LED chip may include multiple LED chips configured to emit different color lights, e.g., red, green, blue lights, but not being limited thereto. Further, by adding cyan or yellow to the red, green, and blue micro LED, it can broaden the color gamut.

FIG. 6 illustrates a top view of a light emitting diode package in accordance with another embodiment of the present invention. The light emitting diode package includes a substrate **102** and at least one LED chip **106** electrically mounted on a top surface of the substrate **102**. The substrate **102** has a top surface with an area **A1** while the LED chip

106 has its light emitting surface with an area **A2**. A ratio of the area **A2** to the area **A1** is preferably equal to or less than 5% to enhance the contrast ratio for the LED package such that the ratio of the side emission to the top emission can be reduced. For example, the ratio of the area **A2** to the area **A1** is 1% in embodiment B; the ratio of the area **A2** to the area **A1** is 5% in embodiments C, D, and E. In embodiments B-E, at least one LED chip **106** may be multiple LED chips, e.g., red, green, and blue LED chips, and the area **A2** may be referred as a total sum area of the light emitting surfaces of all the LED chips. In case the top surface of the substrate **102** is fully covered by the black material layer **108** except the light emitting surface of the LED chip **106**, the ratio of the side emission to the top emission may be further reduced, the contrast ratio may be further enhanced, and extraneous light may not be reflected to cause false lighting.

The ratio of the side emission to the top emission, as discussed in previous embodiments, is reduced to improve crosstalk issues between pixels on the LED display panel.

FIG. 7 illustrates a cross-sectional view of a light emitting diode package **100f** in accordance with another embodiment of the present invention. The LED package **100f** includes a substrate **102f**, at least one micro LED chip **106f** and a transparent material layer **104f**. In this embodiment, the transparent material layer **104f** is a transparent dielectric layer instead of a transparent glue layer, e.g., a transparent resin layer. The transparent dielectric layer may be formed by a chemical vapor deposition process or an atomic layer deposition process, but not being limited thereto. The transparent dielectric layer may be SiO_2 , Al_2O_3 , TiO_2 , Ta_2O_5 , HfO_2 , ZrO_2 , Y_2O_3 , MgF_2 or Si_3N_4 , but not being limited thereto. The transparent dielectric layer, formed by a chemical vapor deposition process or an atomic layer deposition process, may have a thickness ranging from 0.1 micrometer to 10 micrometers, e.g., from 0.1 to 2 micrometers, from 2 to 4 micrometers, from 4 to 6 micrometers, or from 6 to 10 micrometers such that it can be conformal over upper surfaces of the micro LED chip **106f** and the substrate **102f**. In other embodiments, a black material layer, e.g., **108** in FIG. 5, may be added to cover a top surface of the substrate **102f** and expose a light-emitting surface of the at least one micro LED chip **106f**, and be covered by the transparent material layer **104f**.

FIG. 8 illustrates a cross-sectional view of a micro LED chip in accordance with another embodiment of the present invention. The micro LED chip includes a semiconductor stack **120'** and a supporting breakpoint **BP**. The semiconductor stack **120'** includes a first semiconductor layer **121'**, a light emitting layer **122'**, and a second semiconductor layer **123'**. The first semiconductor layer **121'** has a light emitting surface **S1** exposed outside and the light emitting surface **S1** has a rough texture. The light emitting surface **S1** is formed by applying a laser lift off process to a sapphire substrate or patterned sapphire substrate to obtain a laser-lift-off rough pattern. Please see photos in FIGS. 16-17 for a laser-lift-off rough pattern, i.e., uniform round or hexagonal concaves, on a patterned sapphire substrate. Therefore, the micro LED chip is not equipped with the sapphire substrate or patterned sapphire substrate, but equipped with the laser-lift-off rough pattern to enhance light extraction. Specifically, the LED chip may emit a light passed through the light emitting surface **S1**. The supporting breakpoint **BP** is on the light emitting surface **S1**. In some embodiments, the first semiconductor layer **121'** includes doped semiconductor layers **121b'**, **121c'** and an undoped semiconductor layer **121a'**, and the light emitting surface **S1** is on the undoped semiconductor layer **121a'**. The doped semiconductor layers **121b'**,

121c' are between the light emitting layer 122' and the undoped semiconductor layer 121a'. In some embodiments, the LED chip further includes a first conductive pad 191 and a second conductive pad 192. The first conductive pad 191 is electrically connected to the first semiconductor layer 121', and the second conductive pad 192 is electrically connected to the second semiconductor layer 123'. In some embodiments, the LED chip further includes a conductive contact layer 124' between the second conductive pad 192 and the second semiconductor layer 123'. A sacrificial layer 140 is formed on the light emitting surface S1 of the semiconductor stack 120'. The sacrificial layer 140 is etched to form a supporter 142 protruded from a base portion thereof. A carrier substrate 160 is formed over the sacrificial layer 140. In some embodiments, an adhesive layer 170 may be formed between the sacrificial layer 140 and the carrier substrate 160. The carrier substrate 160 can be adhered to the sacrificial layer 140 by the adhesive layer 170 to enhance an adhesion therebetween. A sidewall leakage reduction layer 182 is formed over a sidewall of the semiconductor stack 120' by an atomic layer deposition process. Thus, the sidewall leakage reduction layer 182 reduces sidewall leakage effect and increases external quantum efficiency (EQE). In this embodiment, the sidewall leakage reduction layer 182 may be formed from Al_2O_3 or SiO_2 , but not being limited thereto.

A bottom mirror layer 181 is formed over an outer surface of the sidewall leakage reduction layer 182 to enhance light extraction via the light emitting surface S1. In this embodiment, the bottom mirror layer 181 may be a distributed Bragg reflector (DBR) formed from alternately layers of SiO_2 and TiO_2 , but not being limited thereto.

The semiconductor stack 120' can be separated from the carrier substrate 160, e.g., a sapphire substrate, when the carrier substrate 160 is removed by breaking the supporting breakpoint BP. A detailed process for manufacturing the LED chip can be cross-referenced to the specifications of U.S. application Ser. No. 16/524,202 filed on Jul. 29, 2019.

FIG. 9 illustrates a cross-sectional view of a micro LED chip in accordance with still another embodiment of the present invention. The semiconductor structure in this embodiment is different from that of FIG. 8 in that the supporting breakpoint BP is on a surface of the semiconductor stack 120' that is opposite to the light emitting surface S1. The light emitting surface S1 is formed by applying a laser lift off process to a sapphire substrate or patterned sapphire substrate to obtain a laser-lift-off rough pattern. Therefore, the micro LED chip is not equipped with the sapphire substrate, but equipped with the laser-lift-off rough pattern to enhance light extraction.

In other embodiments, the micro LED chips (106b-106f) may include the light emitting devices (100, 100a, 100b, 100c) as described in the specifications of U.S. application Ser. No. 16/524,165 filed on Jul. 29, 2019. For example, FIG. 14 illustrates the light emitting device 100 as disclosed in FIG. 1 of U.S. application Ser. No. 16/524,165. The light emitting device 100 has a stacked structure 102 and a first insulating layer 104 covering at least side surfaces of the stacked structure 102. The stacked structure 102 includes a p-type semiconductor layer 106, an n-type semiconductor layer 110, and a light emitting layer 108. The n-type semiconductor layer 110 is on the p-type semiconductor layer 106. The light emitting layer 108 is sandwiched between the p-type semiconductor layer 106 and the n-type semiconductor layer 110. In some embodiments, the p-type semiconductor layer 106 is a p-type GaP layer, and the n-type semiconductor layer 110 is an n-type AlGaInP layer. The

stacked structure 102 further includes an n-type electrode 114, an n-type contact layer 112, a p-type electrode 118, an n-type contact pad 116, and a p-type contact pad 120. The n-type electrode 114 is on the n-type semiconductor layer 110. The n-type contact layer 112 is sandwiched between the n-type semiconductor layer 110 and the n-type electrode 114. The n-type contact pad 116 is on the n-type electrode 114. The p-type electrode 118 is on the p-type semiconductor layer 106. The p-type contact pad 120 is on the p-type electrode 118. In some embodiments, the n-type contact layer 112 is an n-type GaAs layer. In some embodiments, the light emitting layer 108 is a multiple quantum well active layer. In an embodiment, the multiple quantum well active layer is formed of alternating layers of a well layer and a barrier layer. In some embodiments, the light emitting layer 108 emits red light and the light emitting device 100 is a red light emitting device. The stacked structure 102 further includes a semiconductor reflector 122 between the light emitting layer 108 and the n-type contact layer 112. In some embodiments, the n-type semiconductor layer 110 has a first portion 110A and a second portion 110B spaced apart from the first portion 110A by the semiconductor reflector 122. The second portion 110B of the n-type semiconductor layer 110 is closer to the n-type contact layer 112 than the first portion 110A of the n-type semiconductor layer 110. A normal projection of the light emitting layer 108 onto the second portion 110B of the n-type semiconductor layer 110 overlaps a normal projection of the semiconductor reflector 122 onto the second portion 110B of the n-type semiconductor layer 110. The semiconductor reflector 122 can reflect light which is generated in the light emitting layer 108. In particular, the semiconductor reflector 122 can prevent the n-type contact layer 112 from absorbing light which is generated in the light emitting layer 108, thereby enhancing the light emission efficiency of the light emitting device 100. In other words, the semiconductor reflector 122 can redirect the light from passing downwards (i.e., in a direction toward the n-type contact layer 112) to passing upwards (in a direction toward the p-type semiconductor layer 106). In some embodiments, the semiconductor reflector 122 is a distributed Bragg reflector (DBR). The semiconductor reflector 122 includes multiple periods. Each period includes at least a first layer 124 and at least a second layer 126. A refractive index of the first layer 124 is different from a refractive index of the second layer 126. The first layer 124 and the second layer 126 of the semiconductor reflector 122 include aluminum in some embodiments. The refractive index of the first layer 124 and the refractive index of the second layer 126 depend on the atomic percentage of the aluminum therein. For example, the first layer 124 of the semiconductor reflector 122 includes $\text{Al}_x\text{Ga}_{1-x}\text{As}$, in which $0 < x < 1$. The second layer 126 of the semiconductor reflector 122 includes $\text{Al}_y\text{Ga}_{1-y}\text{As}$, in which $0 < y < 1$ and y is different from x. That is to say, an atomic percentage of the aluminum in the first layer 124 is substantially different from an atomic percentage of the aluminum in the second layer 126. In some embodiments, the first insulating layer 104 covers a bottom surface of the stacked structure 102 and exposes a bottom part of the n-type contact pad 116 and a bottom part of the p-type contact pad 120. The first insulating layer 104 has a refractive index less than a refractive index of the p-type semiconductor layer 106 such that light extraction efficiency can be improved by the first insulating layer. The p-type semiconductor layer 106 has a top surface 106a facing away from the semiconductor reflector 122. The top surface 106a of the p-type semiconductor layer 106 has irregularities. In other words, the top surface 106a of the p-type semicon-

ductor layer **106** is a rough surface, thereby improving the light extraction efficiency of the light emitting device **100** by reducing loss due to total internal reflection (TIR) between air and the top surface **106a** of the p-type semiconductor layer **106**.

In other embodiments, the micro LED chips (**106b-106f**) may include the light emitting diode structures (**10**, **20**, **30**) as described in the specifications of U.S. application Ser. No. 16/541,132 filed on Aug. 14, 2019. For example, FIG. **15** illustrates a light emitting diode structure as disclosed in FIG. **50** of U.S. application Ser. No. 16/541,132. Process steps for manufacturing the light emitting diode structure can be referred to FIGS. **39** to **49** of U.S. application Ser. No. 16/541,132.

FIGS. **10-11** illustrate steps of manufacturing a light emitting diode display panel using the light emitting diode chip in FIG. **8**. As shown in FIG. **9**, after micro LED chips, e.g., blue LED chips **107B**, green LED chips **107G**, and red LED chips **107R**, are manufactured on their respective substrates (**152**, **154**, **156**), a temporary substrate **150** is used to serially support blue, green, red LED chips (**107B**, **107G**, **107R**) which are first mass transferred from the respective substrates (**152**, **154**, **156**) using stamps **153**. After the first mass transfer, the blue, green, red LED chips (**107B**, **107G**, **107R**) have their electrodes exposed, a pre-test before molding may be performed to screen out mal-functional LED chips. Next, the temporary substrate **150** attached with all blue, green, red LED chips (**107B**, **107G**, **107R**) in proper positions is then turned upside down to perform second mass transfer and mount all blue, green, red LED chips (**107B**, **107G**, **107R**) on a circuit board **158** for further molding with a transparent material layer **162** to be cut along dashed lines and tested. Only functional LED packages **100'** are transferred and attached on a blue tape **200**, and mal-functional LED packages **100'** are screened out. It is noted that each LED package **100'** includes a cut and divided circuit board **158**. Each LED package **100** can be the LED packages (**100b-100f**) as discussed in previous embodiments. As shown in FIG. **11**, functional LED packages **100** are then mounted on a division circuit board **301** and covered by an encapsulation layer **302** to form a divisional display module **300** of a complete LED display panel **400**. In another embodiment, functional LED packages **100'** may be mounted on a division circuit board **301** without an encapsulation layer **302**. When a mini LED package **100** encapsulating at least one micro LED chip has a package size ranging from 100 micrometers to 1000 micrometers, mal-functional packages can easily be tested and screened out before mounting on a division circuit board. The divisional display module **300** with full functional LED packages **100'** can be achieved. In another embodiment, the division circuit board **301** may include a system controller to control LED packages **100'**, e.g., using a system controller to control a pixel controller included in each LED package. The system controller may be mounted on a top surface **301a** of the division circuit board **301** or on a bottom surface **301b** of the division circuit board **301**.

FIGS. **12-13** illustrate steps of manufacturing a light emitting diode display panel using the light emitting diode chip in FIG. **9**. The steps in FIGS. **12-13** are different from steps in FIGS. **10-11** in that the micro LED chips, e.g., blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'** are mass transferred from the respective substrates (**152**, **154**, **156**) directly to the circuit board **158** via a single mass transfer (e.g., using stamps **153**) instead of two times mass transfers. Therefore, a pre-test before molding may not be performed on the LED chips, but the testing after

molding are still available to screen out mal-functional LED packages before mounting on a division circuit board **301**. In this embodiment, each LED package **100** include a set of blue LED chip **107B'**, green LED chip **107G'**, and red LED chip **107R'**. In other embodiments, each LED package may include a pixel controller to control multiple sets of blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'**. For example, as indicated by M4, 4 sets of blue LED chips **107B'**, green LED chips **107G'**, and red LED chips **107R'** as well as a pixel controller may be packaged together to perform a predetermined function, but not being limited thereto. For example, 6, 9 or more sets of R, G, B LED chips may also be packaged together.

In sum, the mini LED package encapsulating at least one micro LED chip as disclosed herein are configured to offer higher luminance, a higher contrast ratio and reduce cross talks. In addition, the mini LED package encapsulating at least one micro LED chip facilitates easily testing the micro LED chips and screening out mal-functional ones before molding into a final display panel.

The present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein. It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

What is claimed is:

1. A divisional display module, comprising:
 - a division circuit board;
 - a plurality of light emitting diode (LED) packages mounted on the division circuit board; and
 - an encapsulation layer covering the LED packages and the division circuit board, wherein each LED package comprises:
 - a substrate having a width;
 - a plurality of micro LED chips electrically mounted on a top surface of the substrate, wherein each micro LED chip has a thickness smaller than 10 micrometers;
 - a black material layer covering the top surface of the substrate and exposing the plurality of micro LED chips, wherein each micro LED chip has the thickness substantially equal to that of the black material layer; and
 - a transparent material layer covering the plurality of micro LED chips, and the transparent material layer having a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4.
2. The divisional display module of claim 1, wherein the black material layer comprises a thickness of 3 micrometers and a reflectivity smaller than 10%.
3. The divisional display module of claim 1, wherein the transparent material layer has a top texture surface.
4. The divisional display module of claim 1, wherein each micro LED chip comprises:
 - a first semiconductor layer having a light emitting surface exposed outside and the light emitting surface has a rough texture;
 - a light emitting layer disposed on the first semiconductor layer;

11

a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and

a supporting breakpoint on the light emitting surface.

5 5. The divisional display module of claim 1, wherein the plurality of micro LED chips include a blue micro LED chip, a green micro LED chip, and a red micro LED chip.

6. The divisional display module of claim 1, wherein the width of the substrate ranges from 100 micrometers to 1000 micrometers.

7. The divisional display module of claim 1, wherein the black material layer is in contact with a sidewall of each micro LED chip.

8. The divisional display module of claim 1, wherein each micro LED chip comprises a light emitting surface having a first area, the top surface of the substrate has a second area, a ratio of the first area to the second area is equal to or smaller than 5%.

9. The divisional display module of claim 1, wherein the thickness of the transparent material layer is smaller than 100 micrometers.

10. The divisional display module of claim 1, wherein the transparent material layer is a transparent dielectric layer or a transparent resin layer.

11. A light emitting diode (LED) display panel, comprising:

a plurality of divisional display modules, wherein each divisional display module comprises:

a division circuit board;

a plurality of LED packages mounted on the division circuit board; and

an encapsulation layer covering the LED packages and the division circuit board, wherein each LED package comprises:

a substrate having a width;

a plurality of micro LED chips electrically mounted on a top surface of the substrate, wherein each micro LED chip has a thickness smaller than 10 micrometers;

a black material layer covering the top surface of the substrate and exposing the plurality of micro LED chips, wherein each micro LED chip has the thickness substantially equal to that of the black material layer; and

12

a transparent material layer covering the plurality of micro LED chips, and the transparent material layer having a thickness, wherein a ratio of the width of the substrate to the thickness of the transparent material layer is equal to or greater than 4.

12. The LED display panel of claim 11, wherein the black material layer comprises a black photoresist having a reflectivity smaller than 10%.

13. The LED display panel of claim 11, wherein the transparent material layer is a transparent dielectric layer or a transparent resin layer.

14. The LED display panel of claim 11, wherein each micro LED chip comprises:

a first semiconductor layer having a light emitting surface exposed outside and the light emitting surface has a rough texture;

a light emitting layer disposed on the first semiconductor layer;

a second semiconductor layer disposed on the light emitting layer, wherein the second semiconductor layer has a type that is different from the first semiconductor layer; and

a supporting breakpoint on the light emitting surface.

15. The LED display panel of claim 11, wherein each LED package further comprises a pixel controller configured to control the plurality of micro LED chips.

16. The LED display panel of claim 15, wherein the division circuit board further comprises a system controller to control the pixel controller included in each LED package.

17. The LED display panel of claim 11, wherein the transparent material layer has a thickness smaller than 100 micrometers.

18. The LED display panel of claim 11, wherein the width of the substrate ranges from 400 micrometers to 1000 micrometers.

19. The LED display panel of claim 13, wherein the transparent dielectric layer comprises SiO_2 , Al_2O_3 , TiO_2 , Ta_2O_5 , HfO_2 , ZrO_2 , Y_2O_3 , MgF_2 or Si_3N_4 .

20. The LED display panel of claim 13, wherein the transparent dielectric layer is formed by a chemical vapor deposition process or an atomic layer deposition process.

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